Project 2

Classifying images of cats and dogs with flipped images

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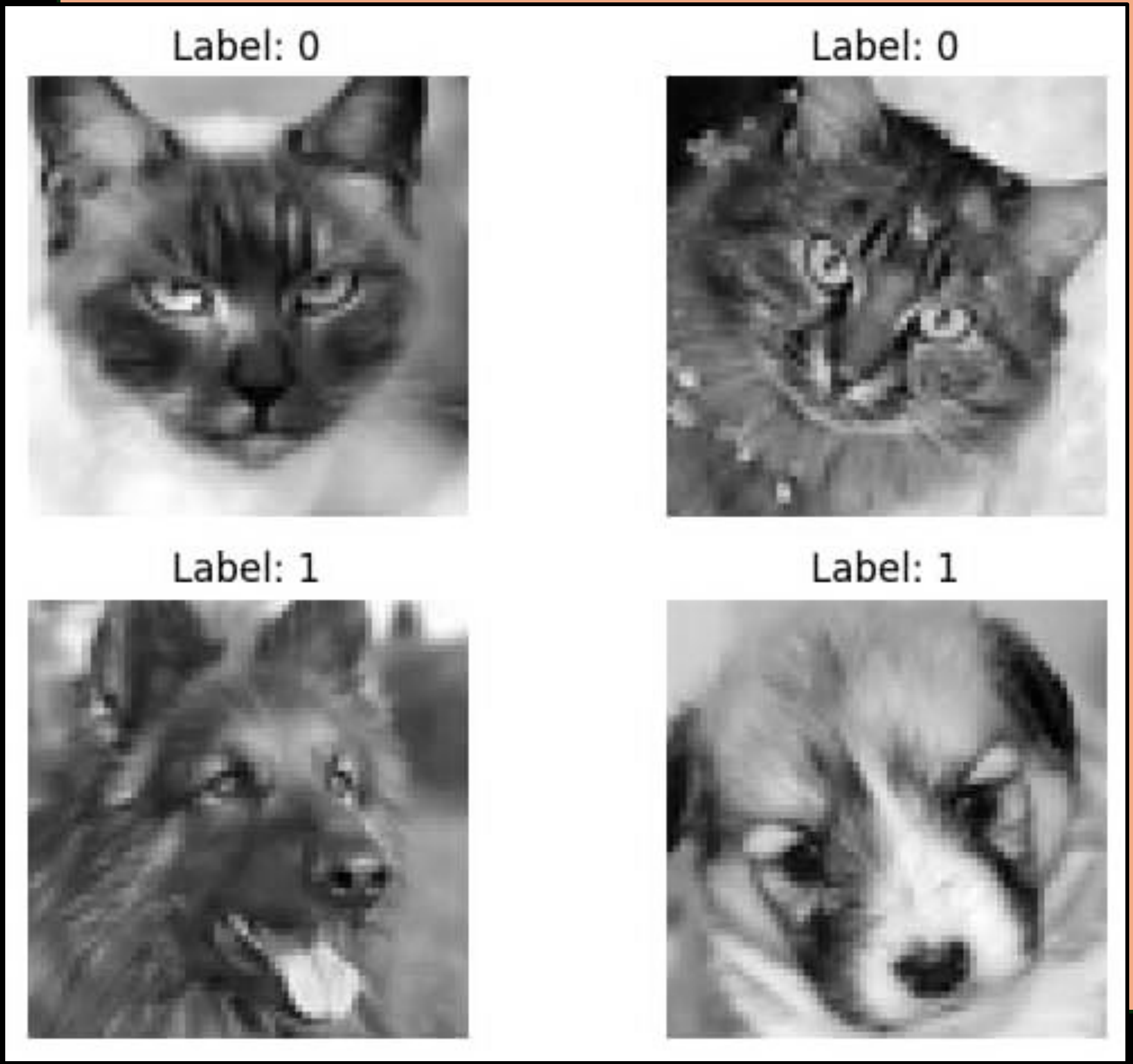
Background

- Cnd large- pictures of cats and dogs
- Binary classification task
  - Cats: 1
  - Dogs: 0
- 10 000 images in total, 50% from each class
- 64x64 = 4096 pixels
- Greyscale



Setup and Analysis

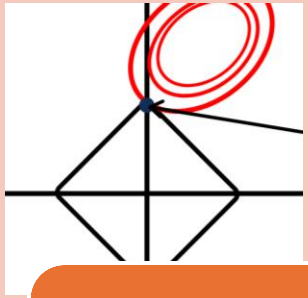
Data exploration

- How does the data look like?
- Balanced?

$$Z = \frac{X - \mu}{\sigma}$$

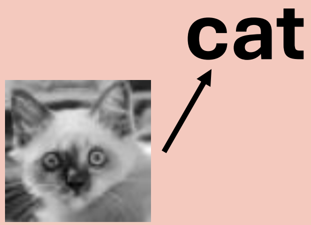
Preprosessing (only for lasso)

- Center and standardise the data.



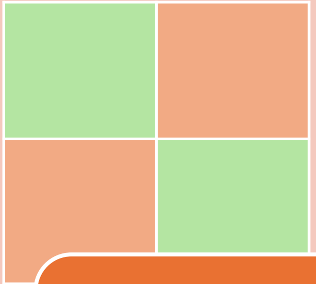
Feature selection

- Lasso
- Random forest



Training and Classifying

- Logistic regression
- CART
- Random Forest



Validating

- Accuracy score
- Cross validation
- Confusion matrix

Hyperparameter tuning

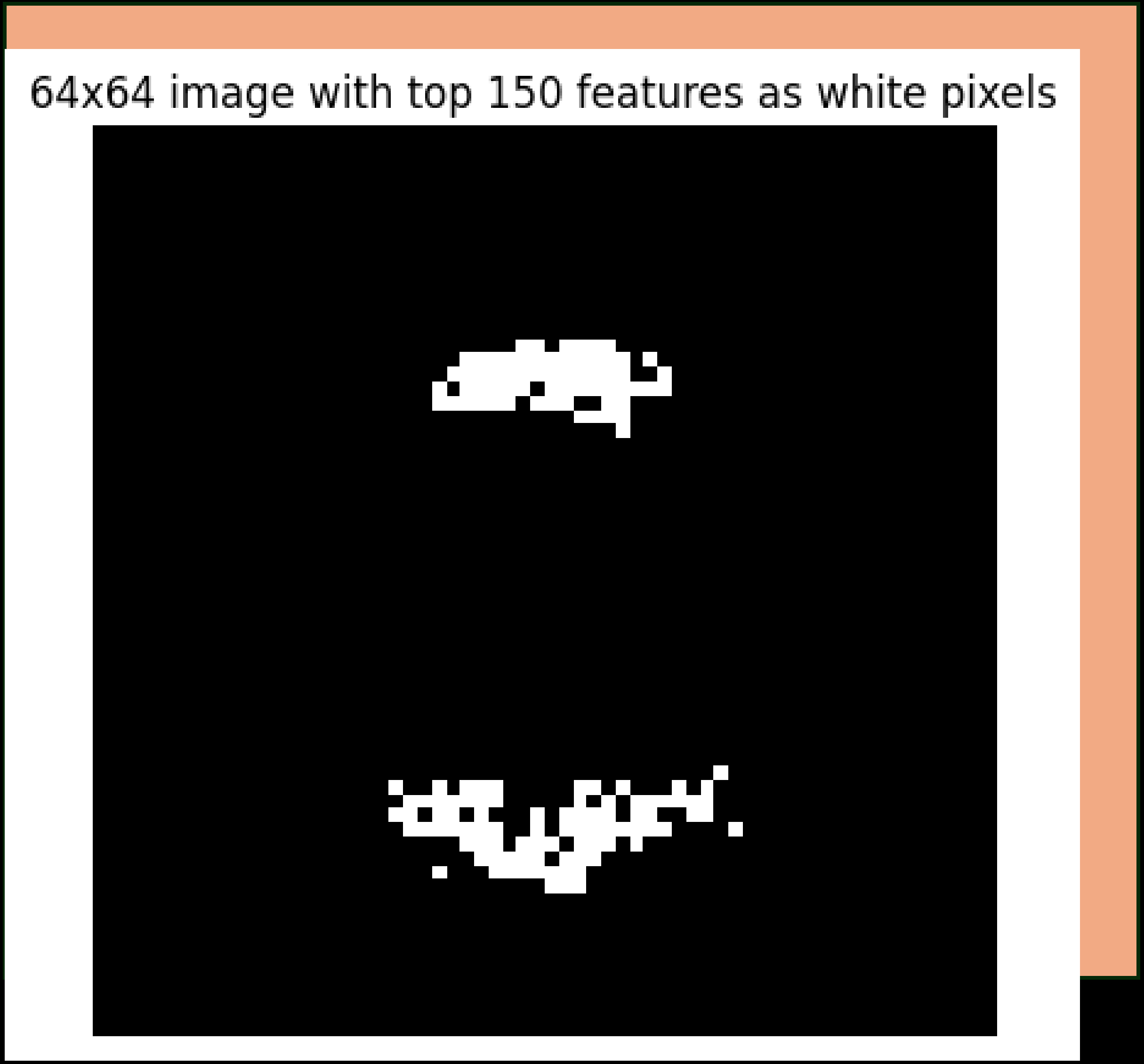
- Grid search
- Using 10 fold cross validation

The hyperparameters for feature selection was tuned before searching for the thresholds

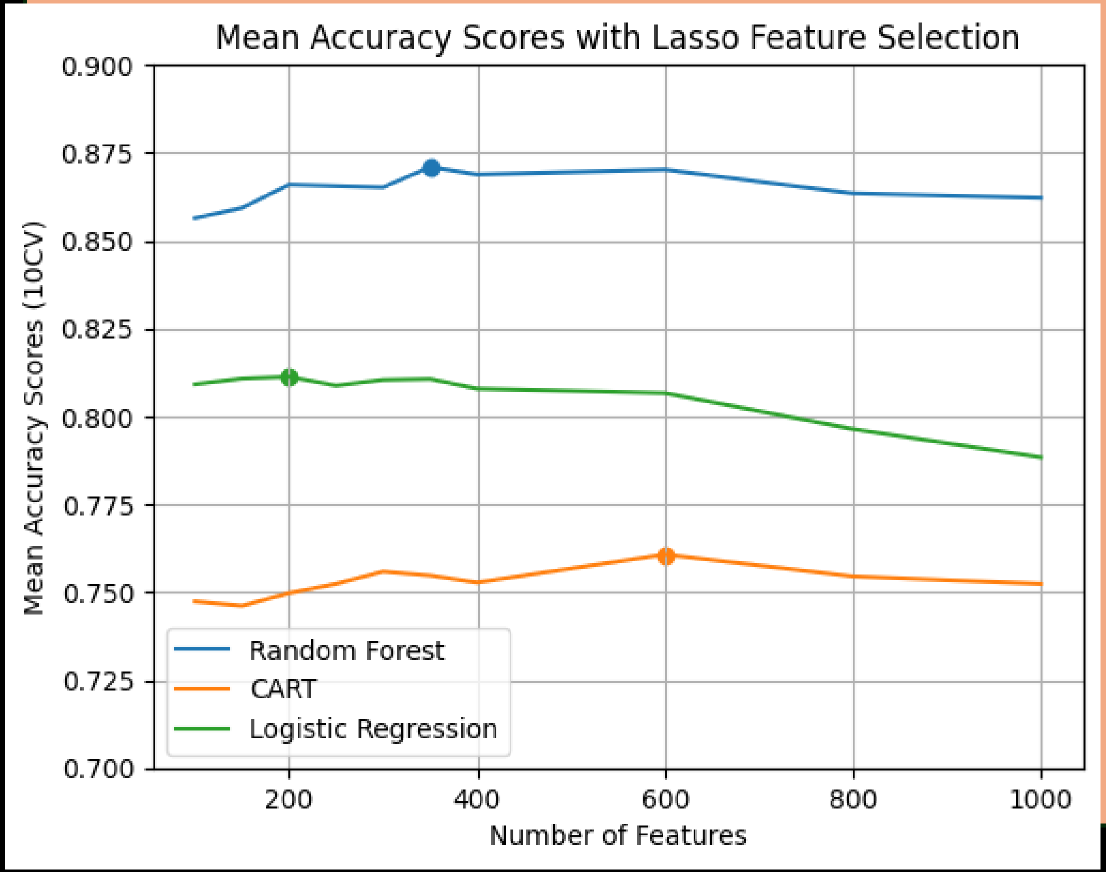
Feature selection

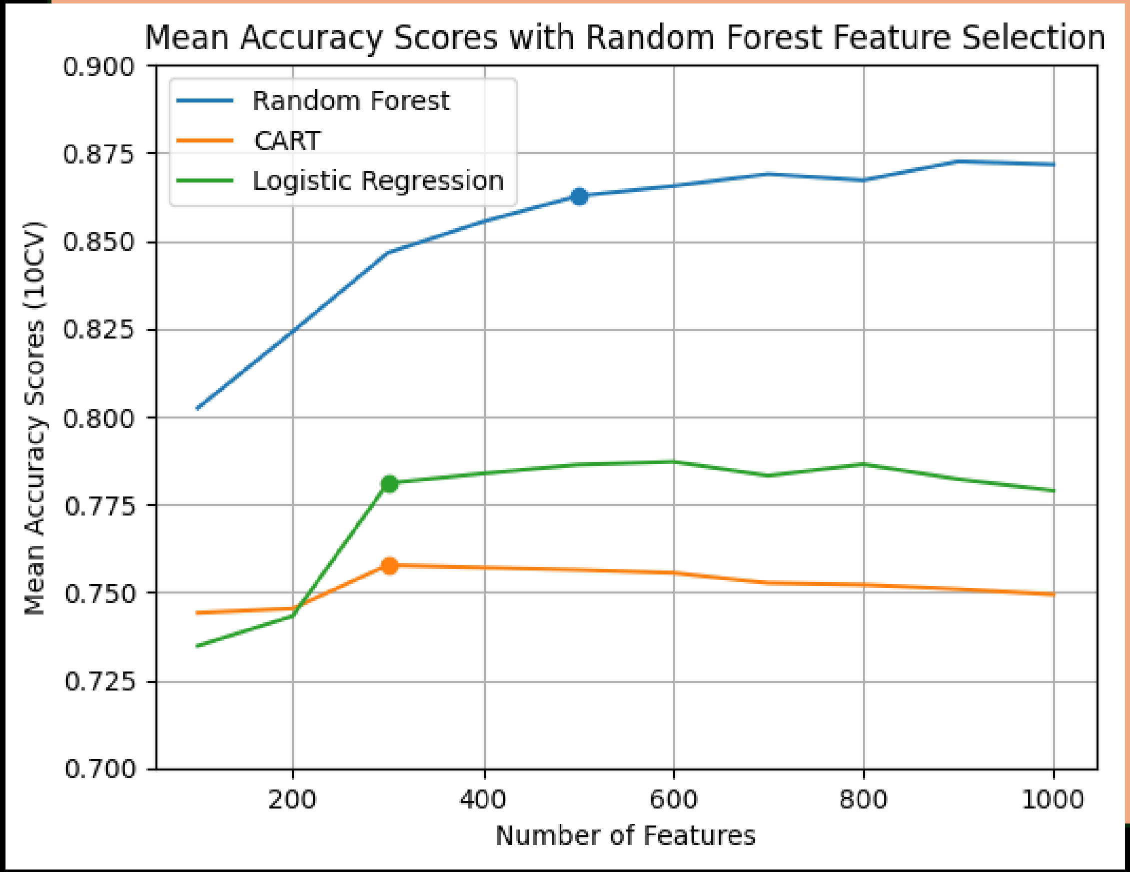
Lasso and RandomForest

- Features (pixels) was selected using Lasso and RandomForest with hyperparameter tuned using GridSearch.
- The optimised models was then used to find the feature threshold indicated with a scatter point.



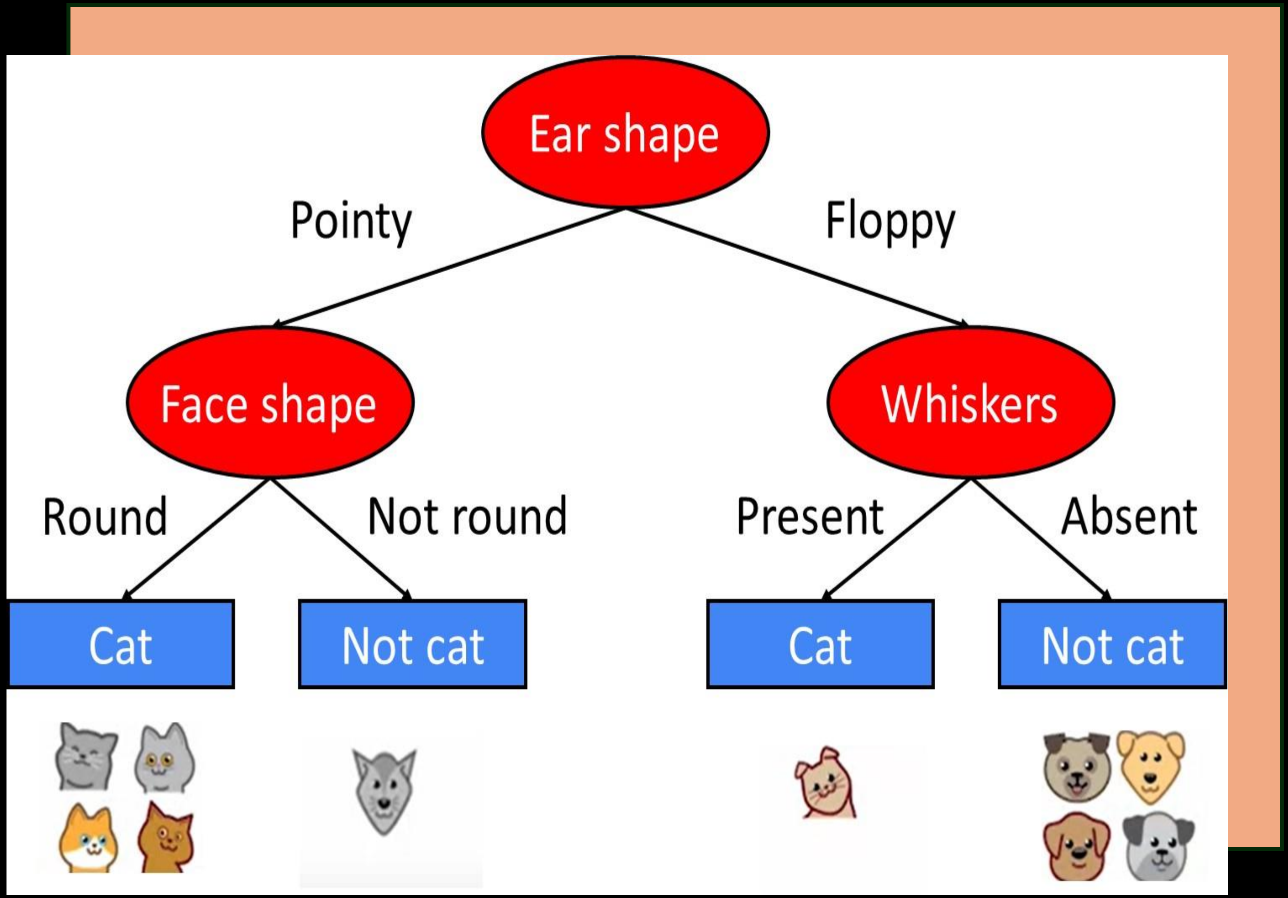
Feature selection





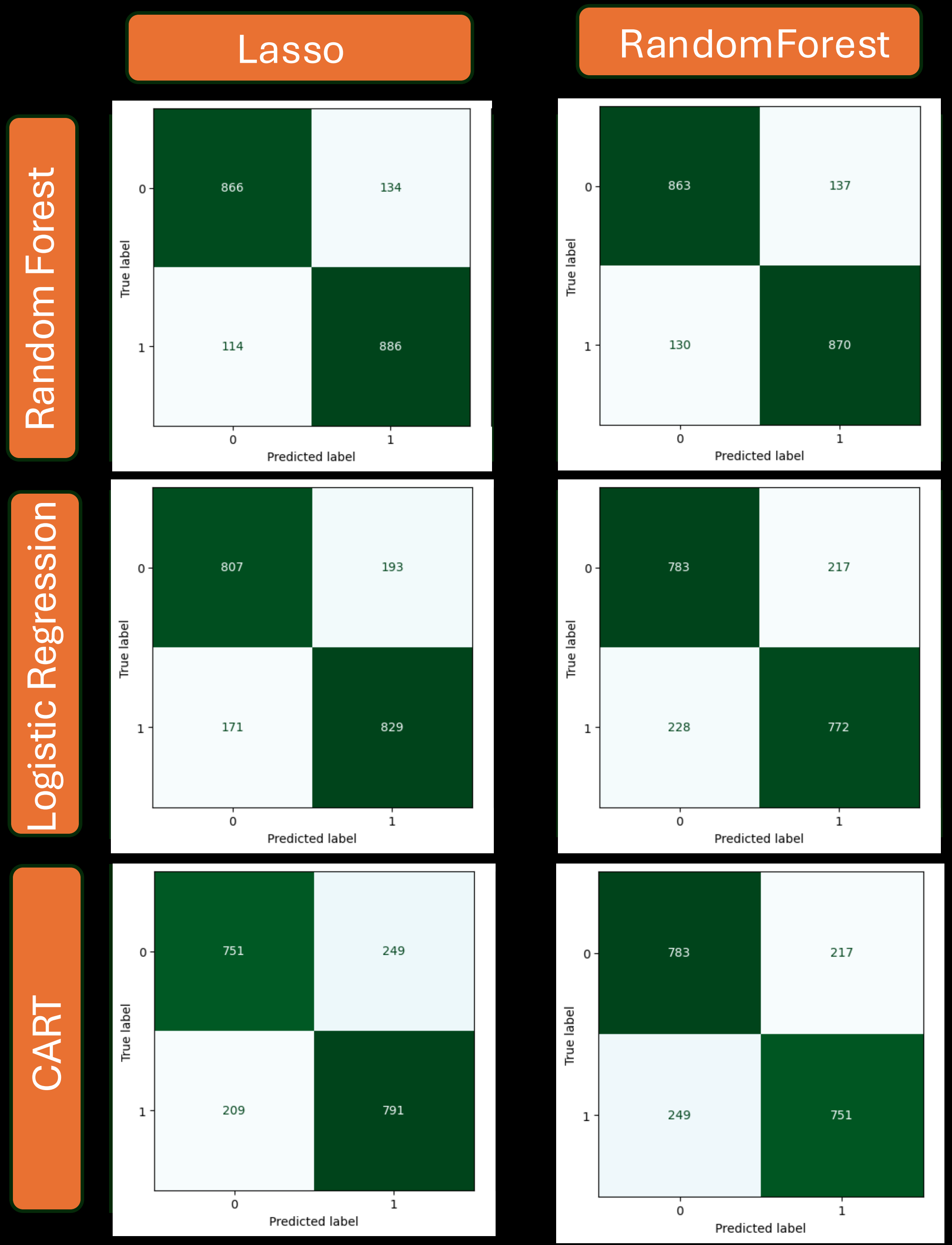
Classifiers

- CART (tree classifier)
- Random Forest
- Logistic regression



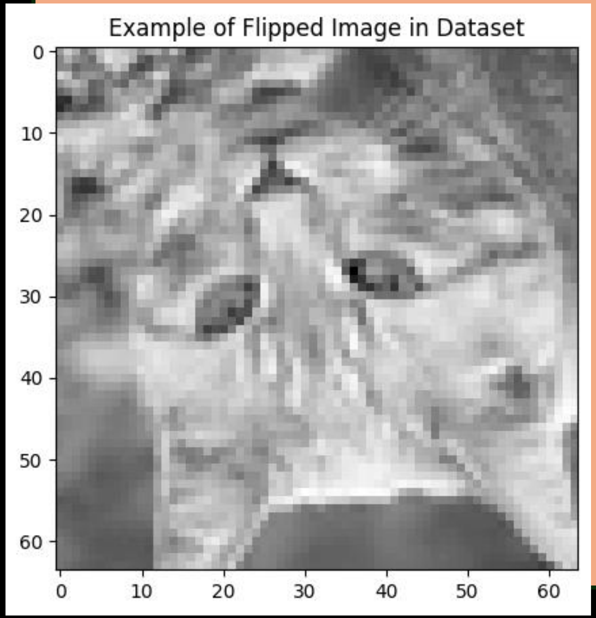
Result part 1

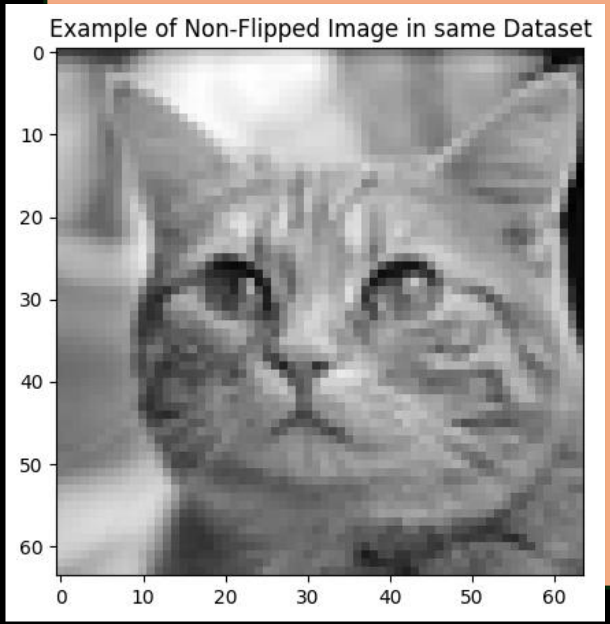
- **Lasso**
  - RandomForest Acc: 0.8760
  - Logistic Regression Acc: 0.8180
  - CART Acc: 0.7575
- **RandomForest**
  - RandomForest Acc: 0.8665
  - Logistic Regression Acc: 0.7775
  - CART Acc: 0.7570



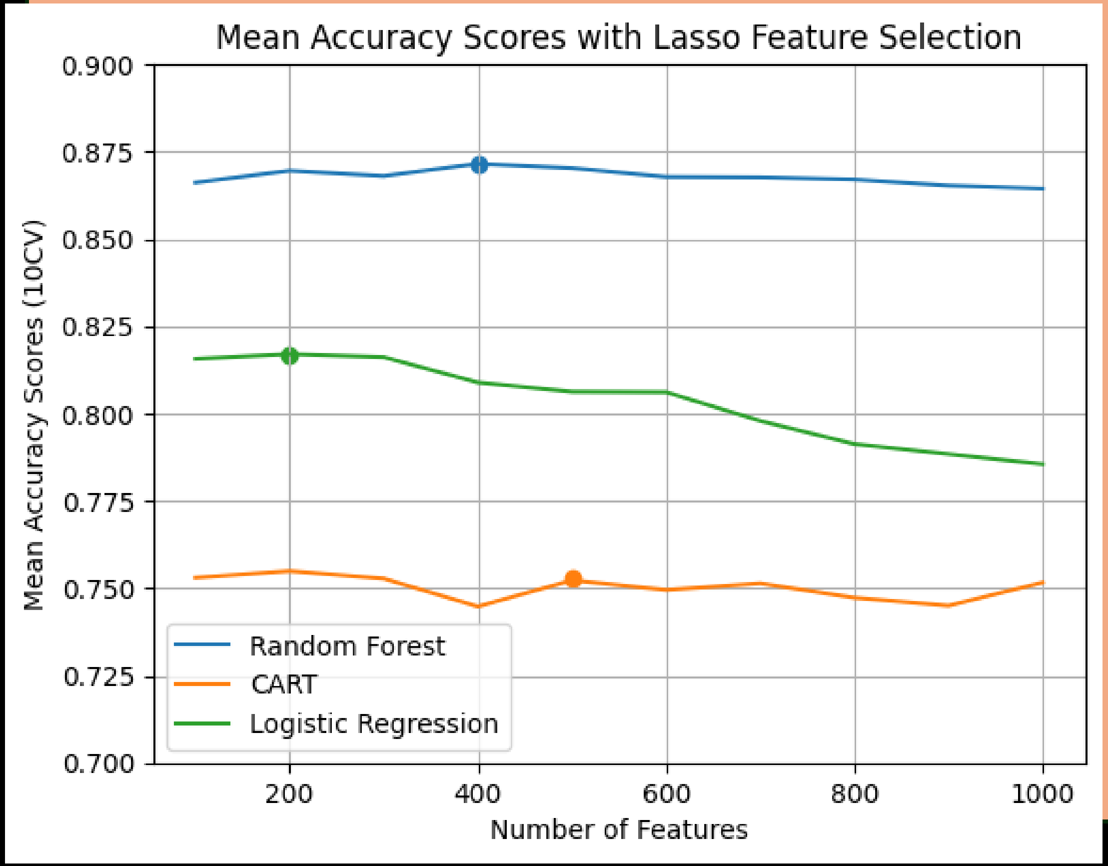
Theme 2: Image Flip

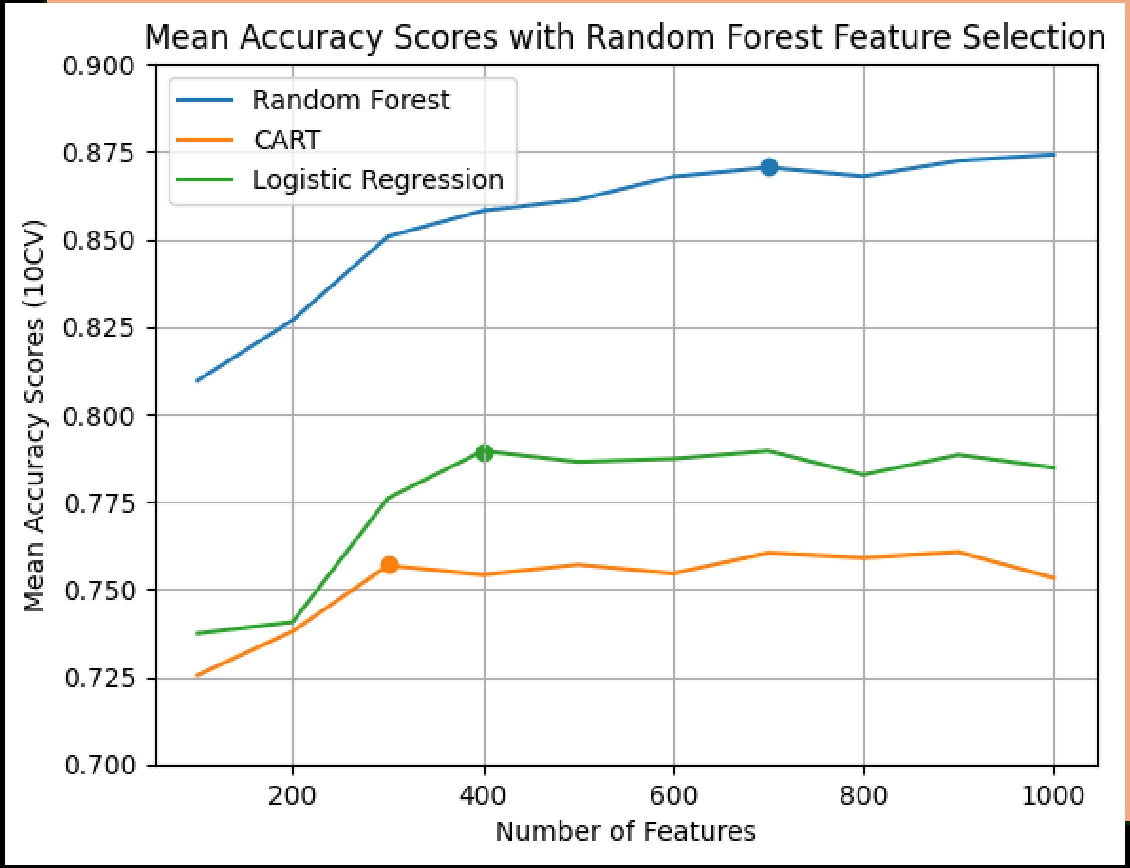
- 50% from each class were flipped
- Then the process from part 1 were repeated
- New thresholds for features





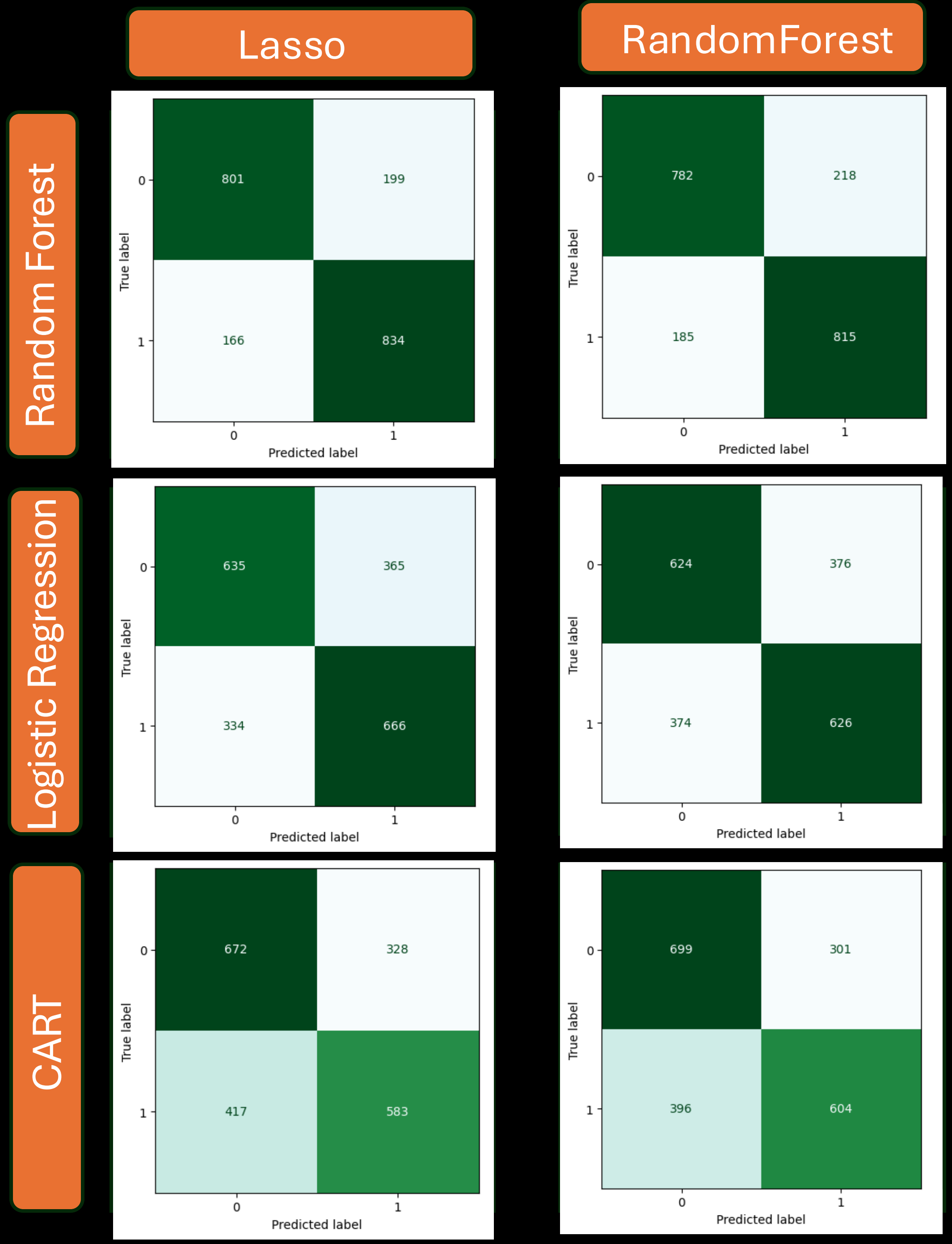
Feature selection





Result part 2

- Lasso
  - RandomForest Acc: 0.8175
  - Logistic Regression Acc: 0.6505
  - CART Acc: 0.6540
- RandomForest
  - RandomForest Acc: 0.7985
  - Logistic Regression Acc: 0.6250
  - CART Acc: 0.6335



Challenges and Surprises

- Computationally heavy for tuning the hyperparameters (several hours)

Take-Home Message

- Complex images makes it difficult to reach accuracy over 0.85
- Random Forest Classifies the best
- Some hundreds features are neccesary for classification